

H-Bridge Based Motor Driver Design

Abdüsselam BOBAROĞLU

19 Mayıs 2026

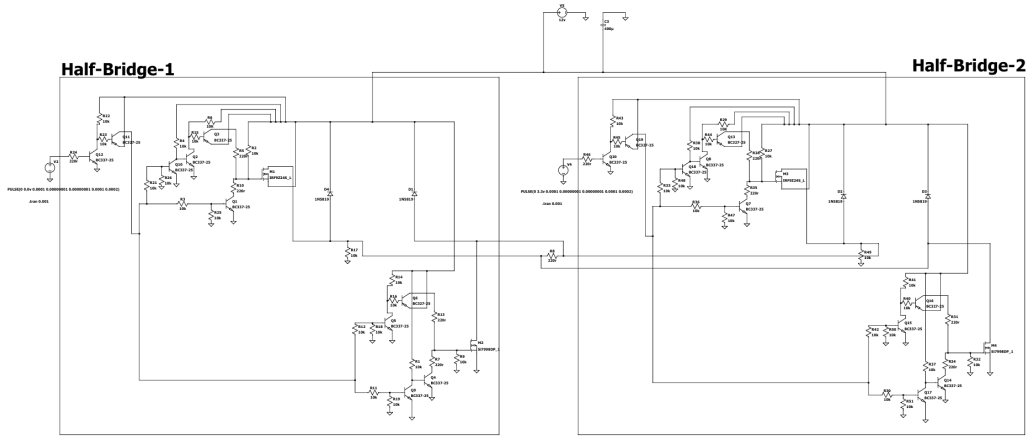
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Bölüm 1

H-Bridge Based Motor Driver Circuit

This study is a continuously developing *living document* that examines the design processes of discrete power electronics topologies. The current report includes the v1.0 stability tests, simulation outputs of the first phase of the project, and the initial thermal/characteristic laboratory findings obtained from the physical prototype.



Şekil 1.1: H-Bridge based motor driver circuit schematic driving a 220 Ω resistive load

Bölüm 2

Experimental Findings and Laboratory Analyses for 10 kHz Triggering Signal

Following the simulation studies, the designed driver topology was physically constructed on a breadboard and tested as a Half-Bridge prototype. The parameters utilized during the experiment and the resulting thermal/characteristic behaviors are summarized below.

A custom-wound coil (inductor) with approximately specified geometric characteristics was utilized as the load of the prototype:

- **Coil Geometry:** 1 mm wire diameter, 108 turns, and 10 cm coil diameter.
- **Input Voltage (Source):** 12 V_{DC} regulated power supply.
- **Control Signal:** Approximately 10 kHz PWM signal (Signal Modulation: 1 s of half-sine wave triggering followed by 1 s of complete cut-off/OFF state).
- **Current Draw:** Approximately 1.0 – 1.5 A in continuous regime.

The specified operating conditions were applied continuously for 15 minutes. As a result of the performed pulse tests, the thermal behaviors of the system components were recorded as follows:

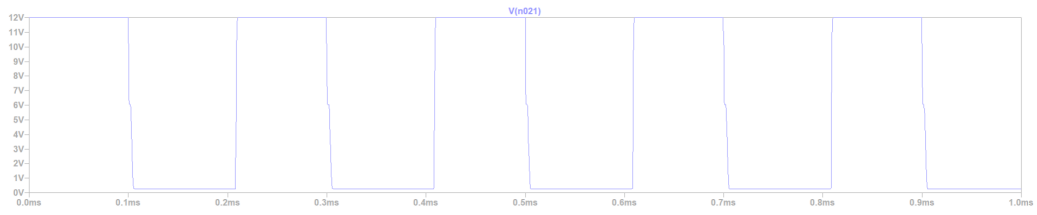
- **PMOS and Driver BJT Stage:** No significant temperature rise or overheating issues were observed in any of the PMOS and BJT transistors. This indicates that the pre-driver stage successfully discharges the PMOS gate capacitance at a frequency of 10 kHz.
- **NMOS Group:** During the tests, the NMOS transistor was found to overheat excessively, reaching a level where it could not be touched by hand.

Engineering Note: *This excessive overheating on the NMOS side is evaluated to be caused by the gate-source threshold voltage (V_{GS}) failing to reach full saturation, forcing the transistor to remain in the linear region and thus exposing it to high $R_{DS(on)}$ conduction losses, or due to reverse recovery losses of the anti-parallel body diodes. As the project progresses, upcoming oscilloscope measurements and optimization results will be integrated into the documentation.*

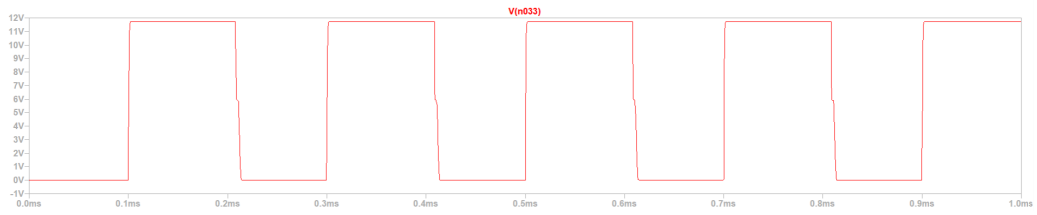
Bölüm 3

Simulation Data for 5 kHz Triggering Signal

3.1 Triggering Signals

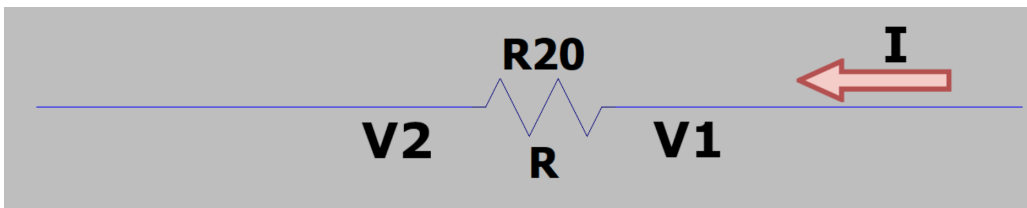


Şekil 3.1: PMOS Gate Terminal Voltage Waveform



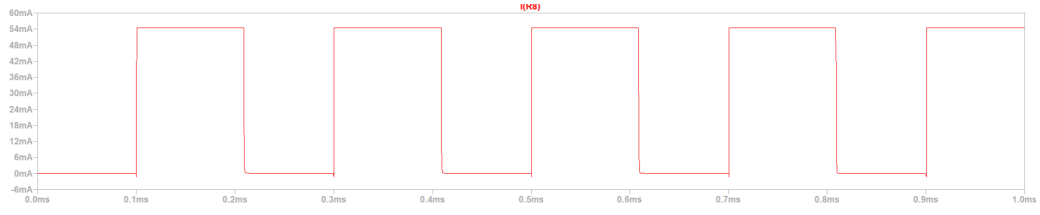
Şekil 3.2: NMOS Gate Terminal Voltage Waveform

3.2 Driven Load: Resistive

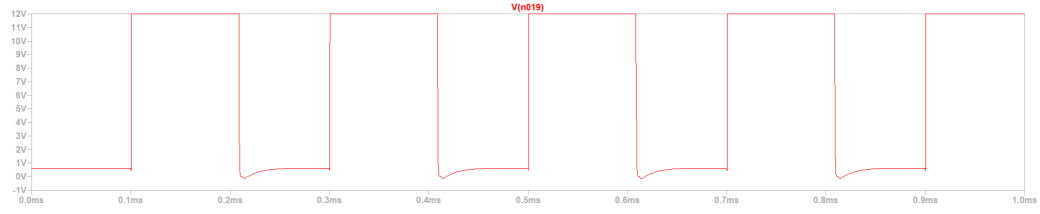


Şekil 3.3: Schematic representation of the measured variables to be plotted

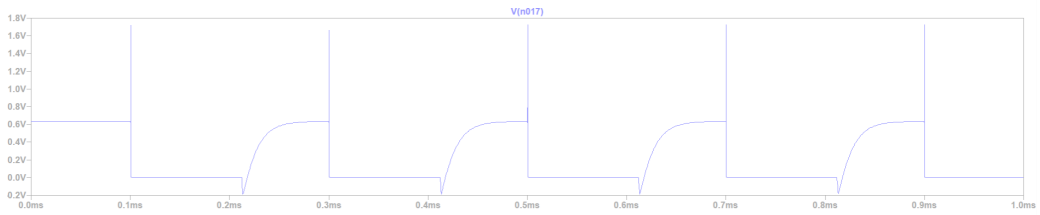
3.2.1 220 Ω Resistor



Şekil 3.4: Current waveform for the 220 Ω resistor

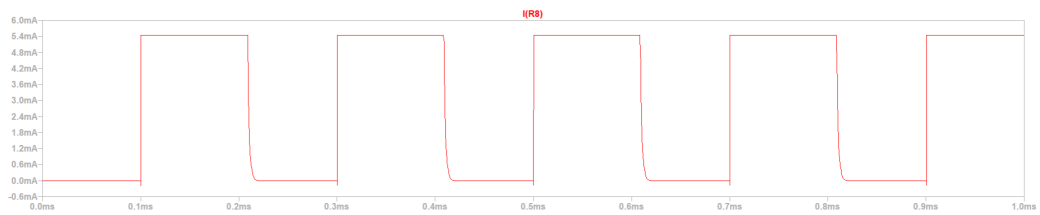


Şekil 3.5: V1 voltage waveform for the 220 Ω resistor

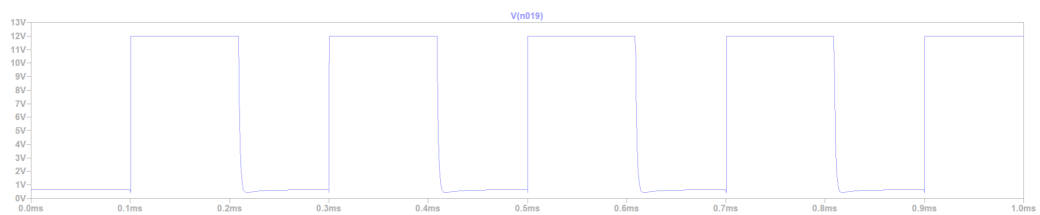


Şekil 3.6: V2 voltage waveform for the 220 Ω resistor

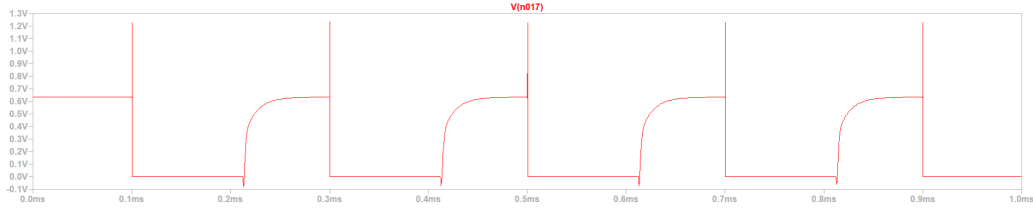
3.2.2 2.2 k Ω Resistor



Şekil 3.7: Current waveform for the 2.2 k Ω resistor

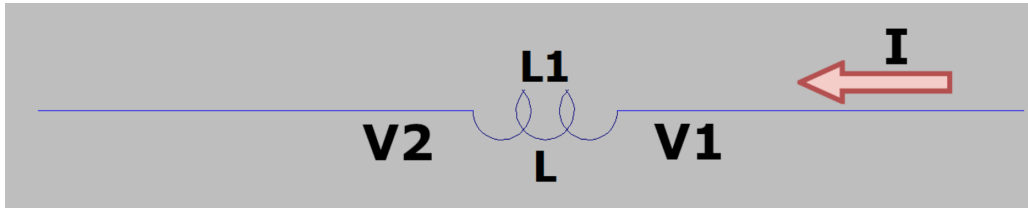


Şekil 3.8: V1 voltage waveform for the 2.2 k Ω resistor



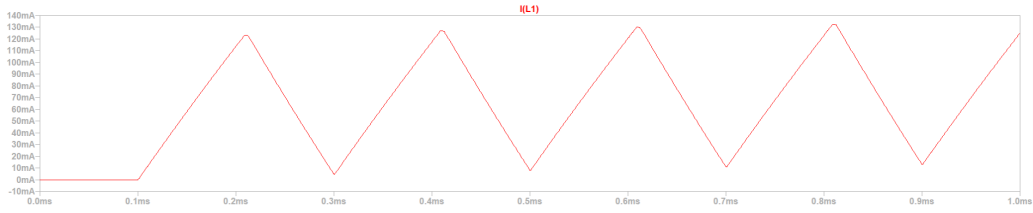
Şekil 3.9: V2 voltage waveform for the 2.2 k Ω resistor

3.3 Driven Load: Inductive

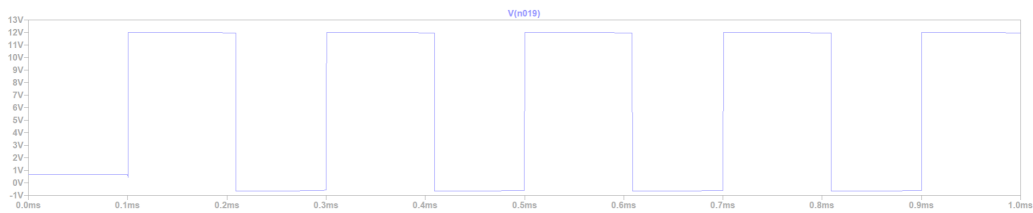


Şekil 3.10: Schematic representation of the measured variables to be plotted

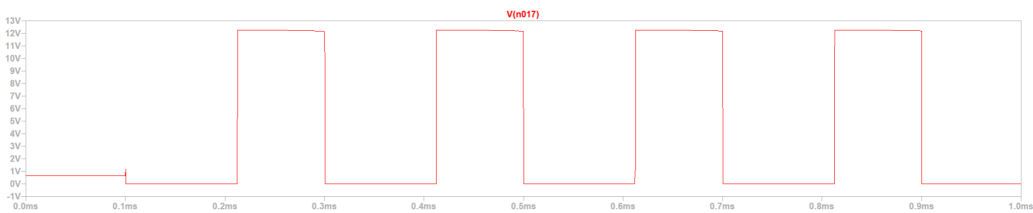
3.3.1 Coil (10 mH Inductance and 10 Ω Internal Resistance)



Şekil 3.11: Current waveform for the 10 mH – 10 Ω coil



Şekil 3.12: V1 voltage waveform for the 10 mH – 10 Ω coil



Şekil 3.13: V2 voltage waveform for the 10 mH – 10 Ω coil